nixlab – Project documentation

As of: 2026-10-19

> This documentation describes a private repo. The domain and internal addresses are replaced with example values: the domain with `homelab.example` (RFC 2606), IPv4 per RFC 5737 (`192.0.2.0/24`, `198.51.100.0/24`, `203.0.113.0/24`), IPv6 per RFC 3849 (`2001:db8::/32`), the tailnet name with `example-tailnet.ts.net`. The real mesh range is a /22 out of RFC 1918. Modules, documentation and build tooling were written together with Claude (Anthropic). License: MIT.

Overview

nixlab is the flake monorepo for the **homelab.example** homelab: five NixOS hosts and a Mac mini running nix-darwin, plus three OpenTofu modules for the NetBird account, the deSEC API tokens and the GitHub repositories with their deploy keys. Everything is operated through a `justfile` from the MacBook; the hosts pull their own configuration from the private GitHub repo and update themselves overnight.

Short answer to the question "*Did you wire NixOS and OpenTofu together somehow?*": **Yes, but deliberately loosely – through files in the repo, not through a shared tool.** OpenTofu writes sops-encrypted files into `hosts/<host>/` that the NixOS modules read at build time, and both sides read the same service list `hosts/grammi/proxy-services.json`. No terranix, no Tofu run from within Nix. Details in the section NixOS & OpenTofu.

Audience: people who can read NixOS flakes, sops-nix and Terraform/OpenTofu.

Repository layout

One flake, `nixosConfigurations` for five hosts plus `darwinConfigurations.zorlock`. Every host gets an explicit module list in `flake.nix`; whatever is missing is missing with the reason in a comment next to it.

Path	Contents
`flake.nix` / `flake.lock`	Inputs, host definitions, packages, `lib.topology`, `checks`
`modules/`	16 shared modules: base, nix, ssh, firewall, networking (ifstate), acme, crowdsec, tailscale, netbird, metrics/metrics-shipping, logging/log-shipping, qemu-guest, deploy, auto-upgrade-git
`hosts/<host>/`	`configuration.nix`, hardware, service modules, sops files (`secrets.yaml`, `netbird.yaml`, `desec.yaml`)
`pkgs/`	Own packages: garage-ui, logchef, paperless-gpt, prometheus-paperless-exporter, ollama-metrics-proxy
`terraform/netbird/`	OpenTofu: NetBird account, policies, DNS zones, reverse proxy services, people/groups (incl. Pocket ID)
`terraform/desec-tokens/`	OpenTofu: narrowly scoped deSEC tokens per host for DNS-01
`terraform/github/`	OpenTofu: the repositories `nixlab` (private) and `nixlab-docs` (public), deploy keys per host
`topology.nix`	Generates the Grafana topology dashboard from the host configs
`justfile`	All operational workflows: deploy, secrets, netbird-*, desec-*

Path	Contents
`docs/`	This project documentation (de/en) and its build, install guides (ursalia, cuckoosegg), drafts of upstream issues
`dnsconfig.js`	Symlink into the separate DNSControl repo – public DNS records live there
`.sops.yaml`	age recipients per host, `admin` (offline recovery) and `workstation` (1Password)

Inputs and their pins

Only `nixpkgs`, `nixpkgs-unstable` and `sops-nix` move with the nightly auto-upgrade (`updateInputs` in `modules/deploy.nix`). Anything that would lock the whole lab out on a broken release is pinned separately and bumped by hand.

Input	Branch / tag	Used by	Nightly?
`nixpkgs`	nixos-26.05	all NixOS hosts	yes
`nixpkgs-unstable`	nixos-unstable	grammi (Grist), ursalia (Paperless 3.x), zorlock (mactop)	yes
`sops-nix`	main	all	yes
`nixpkgs-netbird`	nixos-unstable	NetBird server and all clients	no – server first, then clients
`nixpkgs-pocketid`	nixos-26.05	zummi (pocket-id package)	no
`disko`	main	ursalia (all mounts)	no
`lanzaboote`	v1.1.0	ursalia (Secure Boot)	no
`nixpkgs-darwin`, `nix-darwin`	26.05-darwin	zorlock	no (no auto-upgrade)

`nix flake check` additionally verifies that `hosts/grammi/dashboards/topology.json` matches what `topology.nix` currently generates – a new exporter without `just topology` fails the check.

Modules

The 16 modules under `modules/` carry the configuration shared by all hosts; every file explains in its header comment why it is the way it is. The public version contains them anonymised under `modules/` – as a reference, not directly importable: some read `hosts/<host>/*.yaml` and assume sops-nix. Three are deliberately generic and have options of their own.

Module	Purpose	Options / notes
auto-upgrade-git.nix	Nightly `nixos-rebuild switch`, commits and pushes the new `flake.lock` back to the repo	`services.autoUpgradeGit`: `enable`, `flakeDir`, `flakeAttr`, `updateInputs`, `dates`, `allowReboot`, `rebootWindow`, `deployKeySecret`, `gitAuthorEmail`
log-shipping.nix	journald and nginx logs to VictoriaLogs via Vector	`services.logShipping`: `enable`, `victoriaLogsUrl`
metrics-shipping.nix	vmagent scrapes the local exporters and remote-writes to VictoriaMetrics	`services.metricsShipping`: `enable`, `remoteWriteUrl`, `exporterPorts`
deploy.nix	The lab's policy for auto-upgrade-git: GitHub deploy key, pinned host key, reboot window 04:00–06:00	kept apart from the mechanism so that stays reusable

Module	Purpose	Options / notes
acme.nix	ACME defaults for every TLS host: DNS-01 against deSEC	token from `hosts/<host>/desec.yaml` (from Tofu)
netbird.nix	NetBird client in the mesh, registration via setup key, local metrics	setup key from `hosts/<host>/netbird.yaml` (from Tofu)
tailscale.nix	Shared Tailscale agent	`tailscale serve` stays in the host config
ssh.nix	Who may log in from where: root only via mesh or Tailscale	keys stay per host
firewall.nix	Only 22/80/443 public, tailnet as trusted interface	
crowdsec.nix	CrowdSec engine and firewall bouncer, whitelist for the lab hosts	workarounds for bugs in the nixpkgs modules, documented in the header
networking.nix	ifstate skeleton, systemd-resolved with DNS over TLS, no fallback resolvers	addresses and routes stay in the host config
metrics.nix	Prometheus exporters per host	protected by the firewall instead of bind addresses
logging.nix	Bounds journald, which otherwise grows up to 4 GB	
qemu-guest.nix	QEMU guest agent with an RPC deny list: snapshots yes, shell and file access over the channel no	
nix.nix	Nix daemon, garbage collection, store optimisation	
base.nix	Locale, console, time and tools for debugging	

Hosts & roles

Three rented VMs carry the public services, ursalia and zorlock sit at home, cuckoosegg in the Subraum hackerspace. All of them are reachable in the NetBird mesh as `<host>.peer.homelab.example`; root SSH exists only there.

Host	Platform	Role	Notes
grammi	NixOS, rented VM	NetBird server (`mesh.homelab.example`) + reverse proxy, Traefik, monitoring (VictoriaMetrics/-Logs, Grafana, logchef), Mealie, Grist, PostgreSQL	Central sink for metrics and logs of all hosts
zummi	NixOS, rented VM	Pocket ID (`auth.homelab.example`) – login for all services and NetBird	pocket-id pinned separately
aldrin	NixOS, rented VM	Garage S3 (`s3.aldrin.homelab.example`), garage-ui (`s3-admin`), backup target for restic	Forked garage-ui package with subpath support
ursalia	NixOS, bare metal (ThinkCentre M720q)	Paperless-ngx 3 + paperless-gpt (`dokumente`), Samba for the scanner, routing peer into the home LAN, build host for cuckoosegg	LUKS + btrfs via disko, TPM unlock, Secure Boot (lanzaboote), WLAN (Intel BE200), no CrowdSec
cuckoosegg	NixOS, VM on the Subraum Proxmox	NetBird routing peer into the Subraum networks	Group `subraum-devices` instead of `servers`, no repo/deploy key, deployed by push

Host	Platform	Role	Notes
zorlock	macOS + nix-darwin, Mac mini M4	Ollama on the Apple GPU (`llm`), used by paperless-gpt	None of the `modules/*.nix`; FileVault unlock after reboot via LAN SSH

Services

Services are named by function, not by product. There are three kinds of reachability: public directly, public behind the NetBird reverse proxy with SSO, or mesh only.

Name (`.homelab.example`)	App	Host	Reachable
`auth`	Pocket ID	zummi	public (nginx)
`mesh`	NetBird management/dashboard	grammi	public (Traefik terminates TLS)
`s3.aldrin`	Garage S3 API	aldrin	public (nginx)
`rezepte`	Mealie	grammi	reverse proxy + SSO
`grist`	Grist	grammi	reverse proxy + SSO
`logs`	logchef	grammi	reverse proxy + SSO
`metrics`	Grafana	grammi	reverse proxy + SSO
`vm` / `vl`	VictoriaMetrics / VictoriaLogs UI	grammi	reverse proxy + SSO (via Traefik entrypoint `apps`, port 8090)
`s3-admin`	garage-ui	aldrin	mesh only
`dokumente`	Paperless + paperless-gpt (`/paperless-gpt/`)	ursalia	mesh only; public access via the proxy in progress
`llm`	Ollama (behind Caddy)	zorlock	mesh only

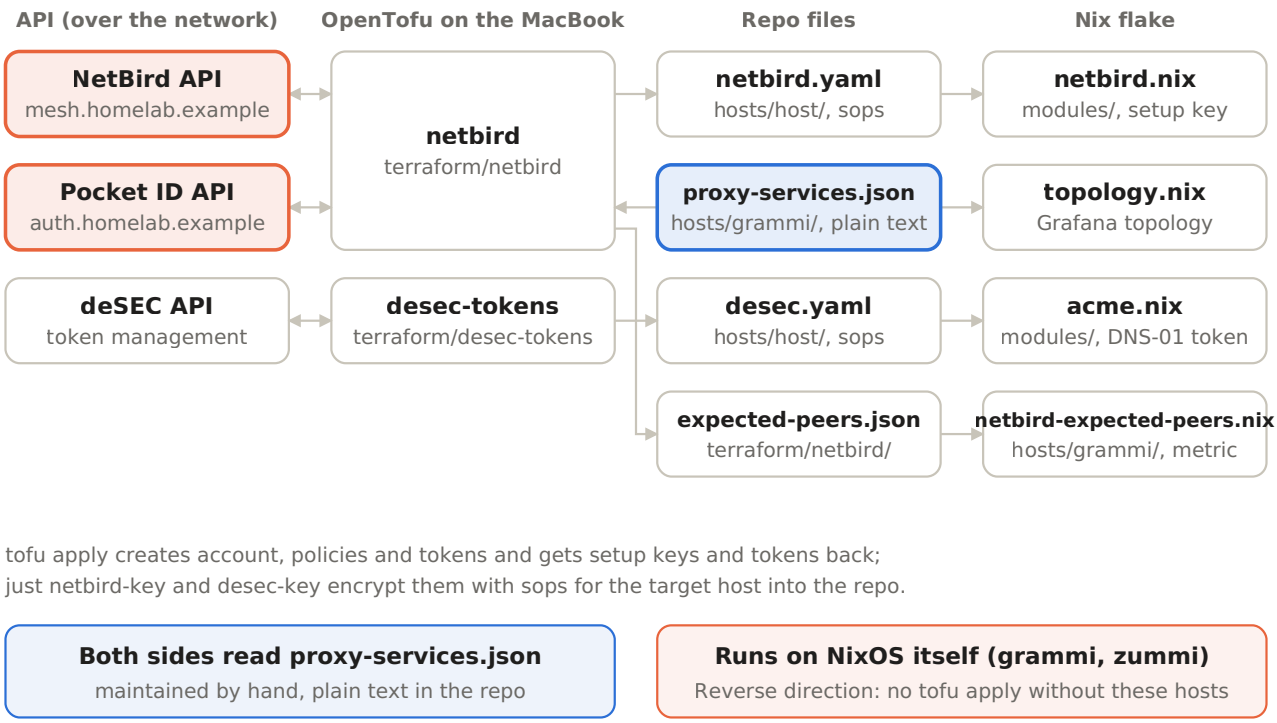
Reverse proxy: Traefik on grammi owns port 443. It terminates `mesh.homelab.example` itself and hands everything else via TCP passthrough with PROXY protocol v2 to `netbird-proxy`, which does TLS and ACME (tls-alpn-01) on its own. Login at the proxy is SSO against Pocket ID, limited to the NetBird groups of the respective app.

Mesh-only names: Every name is its own NetBird DNS zone with one A record pointing at the mesh address (192.0.2.x). Public DNS does not know them; the certificate comes via DNS-01 through deSEC.

Across all Linux hosts: vmagent ships metrics, Vector ships journald and nginx logs to grammi. CrowdSec reads the nginx/Traefik logs on the public hosts. restic backups go to Garage buckets on aldrin, for Paperless additionally append-only to BorgBase.

NixOS & OpenTofu

Yes, the two are wired together – with the repo as the shared interface. OpenTofu manages what exists outside the machines behind an API (NetBird account, deSEC tokens, Pocket ID groups); NixOS manages the machines. The two never talk to each other directly: Tofu only runs on the MacBook through `just`, Nix never evaluates Tofu state.



Tofu never writes to a host directly: every result ends up as a file in the repo and reaches the host only with the next deploy.

The touch points

Direction	File	Created by	Read by	Contents
Tofu → NixOS	`hosts/<host>/netbird.yaml`	`just netbird-key <host>` (`tofu output setup_keys` → `sops encrypt`)	`modules/netbird.nix` (`sopsFile`)	One-time setup key, valid for 24 h
Tofu → NixOS	`hosts/<host>/desec.yaml`	`just desec-key <host>` (`tofu output acme_tokens` → `sops encrypt`)	`modules/acme.nix` (`sopsFile`)	deSEC token for TXT records only, VMs pinned to their IP
Tofu → NixOS	`terraform/netbird/expected-peers.json`	`just netbird-apply` (`tofu output expected_peers`, computed from the policies in `expected-peers.tf`)	`hosts/grammi/netbird-expected-peers.nix` (`builtins.readFile`)	Expected NetBird peers per host as metric `netbird_expected_peers`, next to the real network map in the NetBird dashboard (e.g. aldrin 6, grammi 8)
shared	`hosts/grammi/proxy-services.json`	by hand	`terraform/netbird/reverse-proxy.tf` (`jsondecode(file(...))`) and `topology.nix`	Name → app, port, groups of the reverse proxy services

Direction	File	Created by	Read by	Contents
repo → both	`hosts/<host>/github.yaml` + `github-deploy.pub`	`just github-key <host>` (`ssh-keygen` → `sops encrypt`)	private half: `modules/deploy.nix` (`sopsFile`); public half: `terraform/github` (`file(...)`)	Deploy key for `/etc/nixos`; Tofu only ever sees the public half
NixOS → Tofu	NetBird server on grammi	`hosts/grammi/netbird.nix`	provider via `https://mesh.homelab.example`	The API Tofu runs against in the first place
NixOS → Tofu	Bootstrap tokens	`just netbird-setup-owner`, `netbird-proxy-token` (SSH to grammi, server API)	`terraform/netbird/secrets.yaml` and `hosts/grammi/netbird-proxy.yaml`	Owner PAT, proxy token

The flow for a new host: Tofu creates the setup key, `just netbird-key` writes it encrypted for exactly that host (age key derived from its SSH host key) to `hosts/<host>/netbird.yaml`, commit, deploy within 24 h. The host registers itself on start; Tofu does not manage peers, it looks them up by name via `data "netbird_peer"`.

What is deliberately not wired

- **No shared host list.** Hosts are listed three times: `nixosConfigurations` in `flake.nix`, `local.hosts` in the Tofu modules and `hosts :=` in the `justfile`. A new host has to be added in all three places.
- **No terranix / no Tofu in the Nix build.** Tofu state is local (gitignored, in the main checkout, also from worktrees) because it holds setup keys and tokens in plain text.
- **No provisioning of the machines.** The VMs are created by hand at the provider, ursalia/cuckoosegg following `docs/install-*.md`.
- **Public DNS** lives in DNSControl (its own repo, only linked in via symlink), not in Tofu.

So the line is: whatever Nix can build deterministically from the repo, Nix does. Whatever is created once through an API and returns a secret, Tofu does – and hands it over encrypted as a file.

Secrets, network & access

Secrets (sops-nix, age)

Every host file is encrypted for exactly two recipients: the age key derived from the host's SSH host key and the offline recovery key `admin`. So the MacBook cannot read host secrets – `just secrets <host>` edits them on the host and fetches the result back. Encrypting for a host, on the other hand, works anywhere (only the public key is needed); that is exactly what `netbird-key` and `desec-key` use.

File	Recipients	Contents
`hosts/<host>/secrets.yaml`	host + admin	Deploy key, app passwords, OIDC clients, restic
`hosts/<host>/netbird.yaml`	host + admin	NetBird setup key (from Tofu)
`hosts/<host>/desec.yaml`	host + admin	deSEC token (from Tofu)
`hosts/<host>/github.yaml`	host + admin	GitHub deploy key (from `just github-key`)
`terraform/*/secrets.yaml`	workstation + admin	API tokens for the providers

The workstation key lives in 1Password and is read via `SOPS_AGE_KEY_CMD`, never as a file. `env.sh` in each Tofu module decrypts the tokens only into the shell of the respective recipe.

Network

- **NetBird mesh** (`192.0.2.0/24`, peer names under `peer.homelab.example`): control plane on grammi, login via Pocket ID. Private range (RFC 1918) instead of CGNAT, because tailscaled drops packets from 100.64/10 while both clients run in parallel.
- **Groups:** `servers` (all hosts except cuckoosegg), `subraum-devices`, `admin-workstations` (MacBook), plus person groups from Pocket ID.
- **Home LAN** (`198.51.100.0/24`): routed through ursalia (metric 100), with the two Apple TVs as fallback (group `home-routers`, metric 200, own setup key via `just netbird-router-key`, owned by nobody, in no policy); for group `home-lan`, not for devices that are already at home (posture check `not-at-home`). The Apple TVs need "Quantum-Resistance" on in the app, otherwise no peer (Rosenpass strict) accepts a tunnel to them.
- **ursalia's uplink:** WLAN into the home LAN's main network (Intel BE200, `hosts/ursalia/wifi.nix`): wpa_supplicant, WPA3 with WPA2 fallback, country DE, all bands, passphrase via sops. Its final spot has no cable; eno1 keeps DHCP as a fallback with a worse route metric, so the WLAN stays the primary link when both are up.
- **Subraum:** routed through cuckoosegg for `subraum-access`. cuckoosegg itself may only reach grammi's ingest (TCP 8428/9428) and may only be reached from the MacBook via SSH.
- **Tailscale** still runs in parallel during the migration and serves as break-glass: `NIXLAB_DOMAIN=example-tailnet.ts.net just deploy zummi`.

Access

Only 22/80/443 are open to the public. `modules/ssh.nix` allows root SSH only from mesh or Tailscale addresses; every host has its own login keypair. The mesh is IPv4 only for now; dual stack is listed as an open item.

Deployment & operations

The pull hosts (zummi, aldrin, grammi, ursalia) keep `/etc/nixos` as a checkout of the private GitHub repo and build themselves. cuckoosegg is pushed to, zorlock is switched locally with `darwin-rebuild`.

Host	Manual deploy	Automatic
zummi, aldrin, grammi, ursalia	`just ship <host> "<message>"` = commit, push, `git pull` + `nixos-rebuild switch` on the host	nightly auto-upgrade, reboot window 04:00–06:00
cuckoosegg	`just deploy-push cuckoosegg` (built on ursalia, copied to the target)	ursalia pushes daily at 06:30
zorlock	`just deploy-zorlock` (`darwin-rebuild switch` with sudo)	none

Nightly upgrade

`modules/auto-upgrade-git.nix` updates `nixpkgs`, `nixpkgs-unstable` and `sops-nix`, builds, switches and pushes the new `flake.lock` back to GitHub with the host's deploy key. The GitHub host key is pinned because nobody is watching. A configuration error that only shows up on a cold boot therefore takes the host down at 04:00 – for the VMs there is the provider's console for that.

Important recipes

Recipe	Purpose
`just check <host>`	`git pull` + `nixos-rebuild dry-activate`

Recipe	Purpose
`just secrets <host>` / `pull-secrets <host>`	Edit host secrets on the host and fetch them back
`just topology`	Regenerate the topology dashboard (otherwise `nix flake check` fails)
`just netbird-plan` / `netbird-apply`	Change the NetBird account; apply also rewrites `expected-peers.json` (commit it, deploy grammi)
`just netbird-key <host>`	New setup key for a host, 24 h to deploy
`just netbird-import`	Restore lost Tofu state from the account
`just netbird-renew-token`	Yearly renewal of the service user token
`just github-key <host>`	New GitHub deploy key for a host (key via sops, registration via Tofu)
`just github-plan` / `github-apply`	Change the repositories and deploy keys on GitHub
`just desec-key <host>` / `desec-verify`	Rotate a deSEC token, or check the scopes from every host
`just gc <host>`	Delete store generations older than 7 days
`just docs`	Build this documentation: private PDFs (de/en) and the anonymised public version in `public-docs/` (gitignored clone of `nixlab-docs`), committed and pushed there

Recipes that need sops on the MacBook run with a 1Password sign-in in front, which provides the age key via `SOPS_AGE_KEY_CMD`:

```
eval $(op signin) && export SOPS_AGE_KEY_CMD='op read "op://<vault>/<item>/password"' && just netbird-plan
```

Troubleshooting

- After a `nixos-rebuild`, `systemctl --failed` says nothing about the past – the switch clears failed states. Use `journalctl -u <unit>` instead.
- NetBird down, Tailscale still there: put `NIXLAB_DOMAIN=example-tailnet.ts.net` in front of the recipe.
- Tofu recipes refuse to run against an empty state, so no second `servers` group or set of tokens gets created.
- WLAN on ursalia: `iw dev wlp2s0f0 link` shows band, signal and PHY mode. The router's 5 GHz radio is on a DFS channel with a 300 TU beacon interval, so a single scan can miss it; band, signal, rates and PHY mode also go to VictoriaMetrics (node_exporter `wifi` collector plus `wifi_link_*`), Grafana dashboard "Homelab / WLAN".
- After a network change NetBird can keep a dead P2P path for minutes while still showing `Connected` (seen on ursalia, 2026-10-19). Fix: `systemctl restart netbird-mesh` on the host, via Tailscale if the mesh is down.

Monitoring

Everything ends up on grammi: **vmagent** on every host sends metrics to VictoriaMetrics (`:8428`), **Vector** sends journald and nginx logs to VictoriaLogs (`:9428`), both over the mesh. Grafana (`metrics`) and logchef (`logs`) sit behind the reverse proxy with SSO. Exporters per host come from `modules/metrics.nix`; whatever has no endpoint of its own goes through node_exporter's textfile collector (`/var/lib/prometheus-node-exporter-text-files`), written by small timers or services:

Textfile metric	Host	Source
`tailscale_*`	hosts with Tailscale	`tailscale metrics write`, every 30 s
`fwupd_updates_*`	ursalia	`fwupdmgr get-updates` after each metadata refresh
`wifi_link_*`	ursalia	`iw dev wlp2s0f0 link` (PHY mode, width, MCS, NSS), every 30 s
`netbird_expected_peers`	grammi	`terraform/netbird/expected-peers.json`, baked into the system at build time

The dashboards are JSON in `hosts/grammi/dashboards/`, provisioned from the repo (changes only via commit + deploy of grammi). `topology.json` is generated by `just topology`.

Dashboard	Shows
Hosts Overview	Scrape targets, CPU/memory/disk/network per host, failed units, certificates, backups (btrbk, restic), pending firmware, WLAN band/signal
Topology	Hosts, exporters and data flows as a Graphviz diagram
NetBird	Control plane reachable per host; peers per host as expected (Tofu) / known (network map) / connected P2P / connected relay, plus deviations; connection history, P2P latency, sync and setup times; netbird-server (signal, relay, gRPC, API)
WLAN	ursalia's band (2.4 / 5 / 6 GHz) over time, channel, signal, PHY mode, link rate, traffic, MCS, retries, time associated
Disk Health	SMART/NVMe of ursalia: status, critical warnings, wear, spare, temperature, data written
Paperless	Health (Celery, Redis, DB, index, classifier), documents, inbox, failed tasks, storage, log warnings
paperless-gpt	OCR and suggestion backlog, OCR time per page, LLM requests, durations, tokens, errors; zorlock's GPU and memory
Web & Security	nginx and Traefik requests, status codes, durations; CrowdSec bans, alerts, parser hits
Garage (S3 storage)	aldrin's Garage: cluster health, disk, S3 requests/errors/duration, resync and GC queues
VictoriaMetrics & VictoriaLogs	Ingest rate, disk usage, query duration, active series
zorlock (Mac mini)	mactop: thermal state, CPU/GPU, power, temperatures, memory, fan; Ollama requests, speed, model load times

Expected vs. known peers (NetBird): "expected" is computed by OpenTofu from the policies with the group membership at the last `just netbird-apply` (example: aldrin 6, grammi 8, cuckoosegg 4). "known" is what the management server really distributes. A device that joins later raises known at once, expected only after the next apply + deploy of grammi – on purpose, so a new peer is visible. The model ignores posture checks; see the header of `expected-peers.tf`.

Grafana pitfall: state timelines with `mergeValues` and colour mode "thresholds" turn numbers into threshold buckets ("1+", "< 1") before value mappings are applied. Use colour mode "fixed" and put text and colour into the mappings.

Open items & conventions

Conventions

- One git worktree per topic (`.worktrees/<topic>`), removed and its branch deleted when done. Tofu state always stays in the main checkout.
- Plan first, implement after approval.
- Subdomains by function (`rezepte`, not `mealie`).
- New web services on grammi use PostgreSQL instead of SQLite.
- Dual stack wherever possible.
- Code comments explain the *why*, including the mistakes that led to a decision, with a date.
- Code comments, commit messages and changelogs in English; the project documentation exists in German and English.

Open

☐ Move the NetBird mesh to dual stack (IPv6 overlay)
☐ Finish reverse proxy stage 3: Paperless public under `dokumente` (uploads, paperless-gpt)
☐ Upstream PRs to the NetBird provider (#247 `direct_upstream`/cluster targets, #248 `crowdsec_mode`); then switch the proxy targets on grammi to `localhost`
☐ Remove Tailscale from the servers once nothing depends on it any more
☐ Grafana alerting (contact point still missing) for SMART, fwupd, Paperless, NetBird expected ≠ known, ursalia's WLAN (not associated, stuck on 2.4 GHz, weak signal)
☐ Update notifications for the inputs pinned by hand (NetBird, Pocket ID)
☐ Declarative root passwords via sops
☐ Keep the host list in one place instead of `flake.nix`, the Tofu modules and `justfile`